MIL-W-81241(WP)
5 March 1965

SUPERSEDING
WS 1711
6 November 1962

MILITARY SPECIFICATION

WETTING AGENT

This specification has been approved by the
Bureau of Naval Weapons, Department of the Navy

1. SCOPE

1.1 Scope. This specification covers one type of wetting agent made from sodium dioctyl sulfosuccinate, pentaerythritol dioleate, and glycerol monooleate.

2. APPLICABLE DOCUMENTS

2.1 The following documents, of the issue in effect on date of invitation for bids or request for proposal, form a part of this specification to the extent specified herein.

SPECIFICATIONS

Military

MIL-P-116	Preservation, Methods of
MIL-P-23443(Wep)	Plasticizer, Di(2-ethylhexyl) Adipate (For Use in Rocket Motors)
MIL-R-23444(Wep)	Resin, Plastisol Grade, Polyvinyl Chloride
MIL-S-81238 (Wep)	Sodium Dioctyl Sulfosuccinate
MIL-G-81239 (Wep)	Glycerol Monooleate
MIL-P-81237 (Wep)	Pentaerythritol Dioleate

STANDARDS

Military

MIL-STD-129	Marking for Shipment and Storage

FSC -1337

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(When requesting any of the above documents, give the title and complete designation of the item shown above. Copies of this specification and other unclassified specifications, standards, and publications required by contractors in connection with specific procurement functions may be obtained from the Commanding Officer, Navy Supply Depot (CDS), 5801 Tabor Avenue, Philadelphia, Pennsylvania, 19120.)

3. REQUIREMENTS

3.1 Preproduction samples. Unless otherwise specified in the contract or order, preproduction samples of the wetting agent shall be manufactured using the methods and procedures proposed for the production. The sample shall be tested as specified in Section 4 herein and is for the purpose of determining that, prior to starting production, the contractor's production methods are capable of yielding items that comply with the technical requirements of the contract. After satisfactorily passing all the preproduction tests specified herein, no changes in raw materials and processing of materials for production shall be made without prior written approval of the procuring activity.

3.2 Data requirements. No data is required by this specification or by applicable documents referenced in Section 2 unless specified in the contract or order (see 6.2).

3.3 Material. The material shall be a wetting agent composed of 33.3 ± 1 part by weight of sodium dioctyl sulfosuccinate (MIL-S- 81238 (Wep)), 33.3 ± 1 part by weight of glycerol monooleate (MIL-G- 81239 (Wep)), and 33.3 ± 1 part by weight of pentaerythritol dioleate (MIL-P- 81237 (Wep)).

3.4 Physical and chemical properties

3.4.1 Moisture content. The moisture content shall be a maximum of 1.0 on a weight percent basis.

3.4.2 Acid number. The acid number shall be a maximum of 4.7.

3.4.3 Viscosity. The viscosity in poises shall be a maximum of 100.

3.5 Workmanship. The material shall be uniform in quality and shall be

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free from impurities and other defects that could adversely affect its use.

4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract or purchase order, the supplier is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified, the supplier may utilize his own facilities or any commercial laboratory acceptable to the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.2 Classification of inspections. Inspection of the wetting agent shall be classified as follows:

a. Preproduction inspection (see 4.4)
b. Quality conformance inspection (see 4.5)

4.3 Sampling

4.3.1 Preproduction sample. After mixing each container until the material is homogeneous, a preproduction sample of four 6-ounce samples shall be taken from each container. Combine any two of the four samples to form a composite sample. Repeat with the remaining two samples to form the second composite sample. Each composite sample, manufactured in accordance with 3.1, shall be subjected to the preproduction tests detailed in 4.4 at an activity designated by the procuring activity. Further production of the wetting agent by the supplier, prior to the approval of the preproduction sample, shall be at the supplier's risk.

4.3.2 Quality conformance inspection sampling. Unless otherwise specified, sufficient material shall be taken from each lot to perform the tests as specified in 4.5.

4.3.3 Inspection lot. An inspection lot of the wetting agent shall consist of all material presented for acceptance at one time and produced in a

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single manufacturing batch under homogeneous conditions of manufacture.

4.4 Preproduction inspection. The preproduction sample shall satisfactorily pass the quality conformance inspections detailed in 4.5.

4.5 Quality conformance inspection

4.5.1 Visual examination. Visually examine all containers in the lot to determine compliance with 3.5 and Section 5.

4.5.2 Moisture content test. Determine moisture content by titration with Karl Fischer reagent and standardized water-in-methanol reagent in accordance with the following procedures:

a. Materials. The following specialized equipment and reagents are required:

Aquameter, Model KF-2 with 10 milliliter (ml) burets, Beckman Instruments, Inc., or equivalent. (Use of an equivalent item of equipment will necessitate the use of the manufacturer's recommended procedure for the instrument used.)

Karl Fischer reagent, 3.0 to 5.0 milligrams of water equivalent to 1 ml of reagent, standardized to nearest 0.1 milligram.

Water-in-methanol reagent, 3.0 to 5.0 milligrams of water per ml of reagent, standardized to Karl Fischer reagent.

b. Standardization of water-in-methanol reagent. Neutralize the moisture in 100 ml of anhydrous methanol by adding an excess of Karl Fischer reagent and back titrating with water-in-methanol reagent to the end point. Add 3.0 to 5.0 ml of Karl Fischer reagent from the buret and back titrate to the end point with water-in-methanol reagent. Determine the ratio (R) of Karl Fischer reagent to water-in-methanol reagent.

c. Test procedure. Neutralize the moisture in 100 ml of anhydrous methanol. Add 3 grams of sample, weighed to the nearest milligram, and stir to dissolve. Add an excess of Karl Fischer reagent and stir until the reaction is completed. Back titrate to the end point

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with standardized water-in-methanol reagent. Calculate the percent water as follows:

$$\text{Percent water} = \frac{0.1\ F\ (K - MR)}{W}$$

where:

F = Weight of water equivalent to 1 ml of Karl Fischer reagent, milligrams per ml

K = Volume of Karl Fischer reagent added to sample solution, ml

M = Volume of water-in-methanol reagent to neutralize excess Karl Fischer reagent, ml

R = Ratio of Karl Fischer reagent to water-in-methanol reagent

W = Weight of sample, grams

4.5.3 Acid number test. To 150 ml of absolute ethanol in a 250-ml flask add 2 drops of 1-percent phenolphthalein indicator. Titrate with 0.1 normal sodium hydroxide to a faint pink end point that persists for at least 30 seconds. Add about 10 grams of sample, weighed to the nearest 0.1 milligram and stir to dissolve, warming slightly if necessary. Titrate the sample solution with 0.1 normal sodium hydroxide to the faint pink end point that persists for at least 30 seconds. Calculate the acid number as follows:

$$\text{Acid number} = \frac{56.1 \times V \times N}{W}$$

where:

V = Volume of sodium hydroxide to titrate sample solution, ml

N = Normality of sodium hydroxide

W = Weight of sample, grams

4.5.4 Viscosity test. Weigh 50.0 ± 0.1 grams of di(2-ethylhexyl) adipate (MIL-P-23443(Wep)) into a clean, dry 400-ml beaker on a torsion balance. Add 3.0 ± 0.1 grams of wetting agent and mix thoroughly. Then add 100 ± 1

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grams of polyvinyl chloride resin (MIL-R-23444(Wep)). Place the beaker and contents in an automatic stirrer and stir at 100 to 200 revolutions per minute (rpm) for approximately 5 minutes. Evacuate the mixture in a vacuum desiccator for 30 ± 5 minutes at a pressure of 2 to 4 millimeters of mercury. Then transfer the mixture to a tightly capped bottle and condition in a water bath at 25.0 ± 0.5 degrees centigrade (C) for at least 2 hours. Determine the viscosity of the mixture with a Brookfield viscosimeter, or equivalent, using a number 3 spindle at 6 rpm. Take the reading on the 0-500 scale on the sixth revolution, and calculate the viscosity of the mixture as follows:

$$\text{Viscosity, poise} = \frac{\text{Scale reading} \times 200}{500}$$

4.6 Acceptance criteria. Failure of any preproduction or quality conformance sample to comply with any requirement of this specification shall be cause for rejection of the lot.

5. PREPARATION FOR DELIVERY

5.1 Preservation, packaging and packing. Unless otherwise specified in the contract or order, minimum requirements for preservation, packaging and packing shall be in accordance with MIL-P-116, Method III.

5.2 Marking. Each container shall be marked in accordance with MIL-STD-129. Marking shall include, but not be limited to, the following information:

a. Manufacturer's name and location
b. Material trade name
c. Net weight or volume
d. Lot number, batch number and date of manufacture
e. Shelf life or storage limitations
f. Number and revision letter of this specification

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6. NOTES

6.1 Intended use. The material purchased in accordance with this specification is intended to be used as a wetting agent in rocket motors.

6.2 Ordering data. Procurement documents should specify, but not be limited to, the following information:

a. Title, number and revision letter of this specification
b. Minimum lot size, if applicable
c. Place of delivery
d. Size of container
e. Request for test data
f. Whether preproduction sample is required

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SPECIFICATION ANALYSIS SHEET

Form Approved
Budget Bureau No. 119-R004

INSTRUCTIONS

This sheet is to be filled out by personnel either Government or contractor, involved in the use of the specification in procurement of products for ultimate use by the Department of Defense. This sheet is provided for obtaining information on the use of this specification which will insure that suitable products can be procured with a minimum amount of delay and at the least cost. Comments and the return of this form will be appreciated. Fold on lines on reverse side, staple in corner, and send to preparing activity *(as indicated on reverse hereof)*.

SPECIFICATION

MIL-W-81244(WP) WETTING AGENT

ORGANIZATION *(Of submitter)*	CITY AND STATE

CONTRACT NO.	QUANTITY OF ITEMS PROCURED	DOLLAR AMOUNT
		$

MATERIAL PROCURED UNDER A

☐ DIRECT GOVERNMENT CONTRACT ☐ SUBCONTRACT

1. HAS ANY PART OF THE SPECIFICATION CREATED PROBLEMS OR REQUIRED INTERPRETATION IN PROCUREMENT USE?

A. GIVE PARAGRAPH NUMBER AND WORDING.

B. RECOMMENDATIONS FOR CORRECTING THE DEFICIENCIES.

2. COMMENTS ON ANY SPECIFICATION REQUIREMENT CONSIDERED TOO RIGID

3. IS THE SPECIFICATION RESTRICTIVE?

☐ YES ☐ NO IF "YES", IN WHAT WAY?

4. REMARKS *(Attach any pertinent data which may be of use in improving this specification. If there are additional papers, attach to form and place both in an envelope addressed to preparing activity)*

SUBMITTED BY *(Printed or typed name and activity)*	DATE

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Bureau of Naval Weapons
Washington, D. C. 20360

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